

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{1}{288} \frac{1}{m_s^2} s_Y^2 (\overline{y_e^{i4p}} (y_e^{i3p} (-7g_1^2 + g_2^2) \delta_{i1i2} - 2g_2^2 y_e^{i1p} \delta_{i2i3}) + \right. \\ & \quad \left. \overline{y_e^{i2p}} (-2y_e^{i3p} (18c_Y^2 \overline{y_e^{i4r}} y_e^{i1r} + g_2^2 \delta_{i1i4}) + y_e^{i1p} (-7g_1^2 + g_2^2) \delta_{i3i4}) \right) + \\ & \frac{1}{18} g_2^4 (2\delta_{i1i4} \delta_{i2i3} - \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,0}[m_2] + \frac{1}{12} g_2^4 (2\delta_{i1i4} \delta_{i2i3} - \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,-1}[m_2] + \\ & \frac{4}{45} g_2^4 (-2\delta_{i1i4} \delta_{i2i3} + \delta_{i1i2} \delta_{i3i4}) \text{LF}_{5,-2}[m_2] + \\ & \frac{1}{54} \sum_p g_1^4 \text{LF}_{3,0}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \frac{5}{144} \sum_p g_1^4 \text{LF}_{4,-1}[m_d^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{135} \sum_p g_1^4 \text{LF}_{5,-2}[m_d^p] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} \sum_p g_1^4 \text{LF}_{3,0}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{5}{48} \sum_p g_1^4 \text{LF}_{4,-1}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \frac{2}{45} \sum_p g_1^4 \text{LF}_{5,-2}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{36} \sum_p (2g_2^4 \delta_{i1i4} \delta_{i2i3} + (g_1^4 - g_2^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,0}[m_l^p] - \\ & \frac{5}{96} \sum_p (2g_2^4 \delta_{i1i4} \delta_{i2i3} + (g_1^4 - g_2^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,-1}[m_l^p] + \\ & \frac{1}{45} \sum_p (2g_2^4 \delta_{i1i4} \delta_{i2i3} + (g_1^4 - g_2^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{5,-2}[m_l^p] + \\ & \frac{1}{108} \sum_p (18g_2^4 \delta_{i1i4} \delta_{i2i3} + (g_1^4 - 9g_2^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,0}[m_q^p] - \\ & \frac{5}{288} \sum_p (18g_2^4 \delta_{i1i4} \delta_{i2i3} + (g_1^4 - 9g_2^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,-1}[m_q^p] + \\ & \frac{1}{135} \sum_p (18g_2^4 \delta_{i1i4} \delta_{i2i3} + (g_1^4 - 9g_2^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{5,-2}[m_q^p] + \\ & \frac{2}{27} \sum_p g_1^4 \text{LF}_{3,0}[m_u^p] \delta_{i1i2} \delta_{i3i4} - \frac{5}{36} \sum_p g_1^4 \text{LF}_{4,-1}[m_u^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{135} \sum_p g_1^4 \text{LF}_{5,-2}[m_u^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{12} s_Y^2 (g_1^2 \overline{y_e^{i4p}} y_e^{i3p} \delta_{i1i2} + \overline{y_e^{i2p}} (-3c_Y^2 \overline{y_e^{i4r}} y_e^{i1r} y_e^{i3p} + g_1^2 y_e^{i1p} \delta_{i3i4})) \text{LF}_{1,2}[m_\Phi] + \\ & \frac{1}{48} s_Y^2 (\overline{y_e^{i4p}} (y_e^{i3p} (g_1^2 + g_2^2) \delta_{i1i2} - 2g_2^2 y_e^{i1p} \delta_{i2i3}) + \\ & \quad \overline{y_e^{i2p}} (2y_e^{i3p} (3s_Y^2 \overline{y_e^{i4r}} y_e^{i1r} - g_2^2 \delta_{i1i4}) + y_e^{i1p} (g_1^2 + g_2^2) \delta_{i3i4})) \text{LF}_{2,1}[m_\Phi] + \\ & \frac{1}{144} (8g_2^4 \delta_{i1i4} \delta_{i2i3} + 3s_Y^2 \overline{y_e^{i4p}} (-y_e^{i3p} (g_1^2 + g_2^2) \delta_{i1i2} + 2g_2^2 y_e^{i1p} \delta_{i2i3}) + \\ & \quad 4(g_1^4 - g_2^4) \delta_{i1i2} \delta_{i3i4} + 3s_Y^2 \overline{y_e^{i2p}} (2g_2^2 y_e^{i3p} \delta_{i1i4} - y_e^{i1p} (g_1^2 + g_2^2) \delta_{i3i4})) \text{LF}_{3,0}[m_\Phi] + \\ & \frac{1}{96} (-10g_2^4 \delta_{i1i4} \delta_{i2i3} + 5(-g_1^4 + g_2^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,-1}[m_\Phi] + \\ & \frac{1}{45} (2g_2^4 \delta_{i1i4} \delta_{i2i3} + (g_1^4 - g_2^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{5,-2}[m_\Phi] + \\ & \frac{1}{36} (2g_2^4 \delta_{i1i4} \delta_{i2i3} + (g_1^4 - g_2^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,0}[\tilde{\mu}] + \\ & \frac{1}{24} (2g_2^4 \delta_{i1i4} \delta_{i2i3} + (g_1^4 - g_2^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,-1}[\tilde{\mu}] + \\ & \frac{1}{45} (-4g_2^4 \delta_{i1i4} \delta_{i2i3} + 2(-g_1^4 + g_2^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{5,-2}[\tilde{\mu}] + \\ & \frac{1}{192} g_1^2 (2g_2^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,1,0}[m_1, m_l^{i1}] + \\ & \frac{1}{192} g_1^2 (2g_2^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,2,-1}[m_1, m_l^{i1}] + \\ & \frac{1}{96} g_1^2 (-2g_2^2 \delta_{i1i4} \delta_{i2i3} + (-g_1^2 + g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,1,-1}[m_1, m_l^{i1}] + \\ & \frac{1}{192} g_1^2 (2g_2^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,1,-2}[m_1, m_l^{i1}] + \\ & \frac{1}{192} g_1^2 (2g_2^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,1,0}[m_1, m_l^{i2}] + \\ & \frac{1}{192} g_1^2 (2g_2^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,2,-1}[m_1, m_l^{i2}] + \\ & \frac{1}{96} g_1^2 (-2g_2^2 \delta_{i1i4} \delta_{i2i3} + (-g_1^2 + g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,1,-1}[m_1, m_l^{i2}] + \\ & \frac{1}{192} g_1^2 (2g_2^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,1,-2}[m_1, m_l^{i2}] + \\ & \frac{1}{192} g_1^2 (2g_2^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,1,0}[m_1, m_l^{i3}] + \\ & \frac{1}{192} g_1^2 (2g_2^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,2,-1}[m_1, m_l^{i3}] + \\ & \frac{1}{96} g_1^2 (-2g_2^2 \delta_{i1i4} \delta_{i2i3} + (-g_1^2 + g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,1,-1}[m_1, m_l^{i3}] + \\ & \frac{1}{192} g_1^2 (2g_2^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,1,-2}[m_1, m_l^{i3}] + \\ & \frac{1}{192} g_1^2 (2g_2^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,1,0}[m_1, m_l^{i4}] + \\ & \frac{1}{192} g_1^2 (2g_2^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,2,-1}[m_1, m_l^{i4}] + \\ & \frac{1}{96} g_1^2 (-2g_2^2 \delta_{i1i4} \delta_{i2i3} + (-g_1^2 + g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,1,-1}[m_1, m_l^{i4}] + \\ & \frac{1}{192} g_1^2 (2g_2^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,1,-2}[m_1, m_l^{i4}] + \\ & \frac{1}{192} (-2g_2^4 \delta_{i1i4} \delta_{i2i3} + g_2^2 (3g_1^2 + g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,1,0}[m_2, m_l^{i1}] + \\ & \frac{1}{192} (-2g_2^4 \delta_{i1i4} \delta_{i2i3} + g_2^2 (3g_1^2 + g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,2,-1}[m_2, m_l^{i1}] + \\ & \frac{1}{96} (-10g_2^4 \delta_{i1i4} \delta_{i2i3} + g_2^2 (-3g_1^2 + 5g_2^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,1,-1}[m_2, m_l^{i1}] + \\ & \frac{1}{64}$$